

Can LLM Agent Skills Close the Model Size Gap in Threat Hunting?

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Project Overview

Large Language Models (LLMs) demonstrate strong general capabilities yet consistently fall short on expert-level tasks *where accuracy is non-negotiable*.

Wazuh, the open-source SIEM used in this study, turns raw endpoint activity into structured alerts that an analyst – or an LLM – must correctly interpret to reconstruct a real attack.

The Challenge

Cybersecurity **threat hunting** demands expert reasoning across hundreds of thousands of alerts *without guidance*.

We propose **structured skill integration** – where *Agent Skills*, reusable, human-readable domain knowledge documents, are passed into the LLM's prompt at inference time to guide its reasoning – as a lightweight alternative to retraining or RAG.

Research Questions

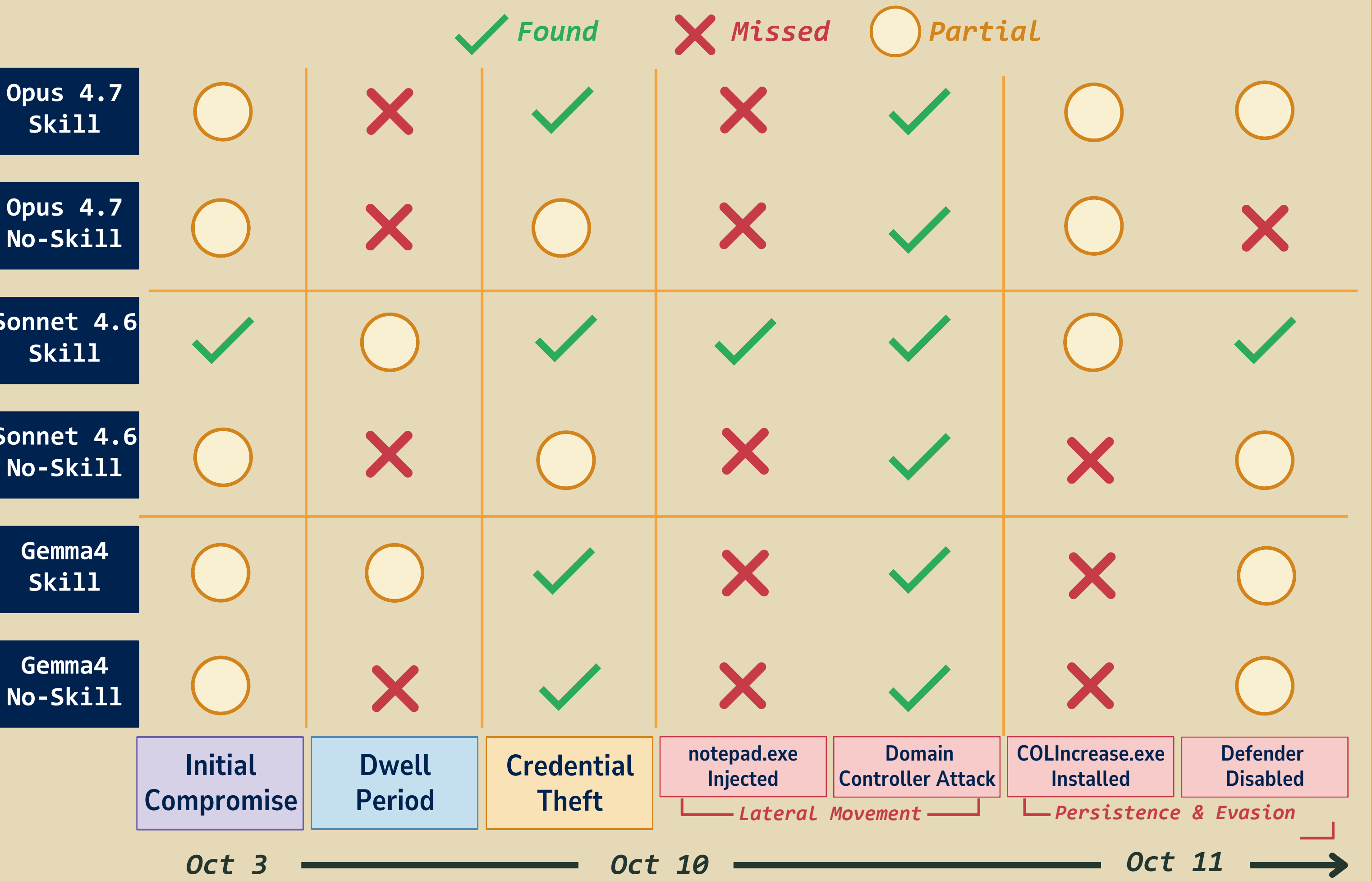
- 1

Does skill integration improve performance holding model constant?
- 2

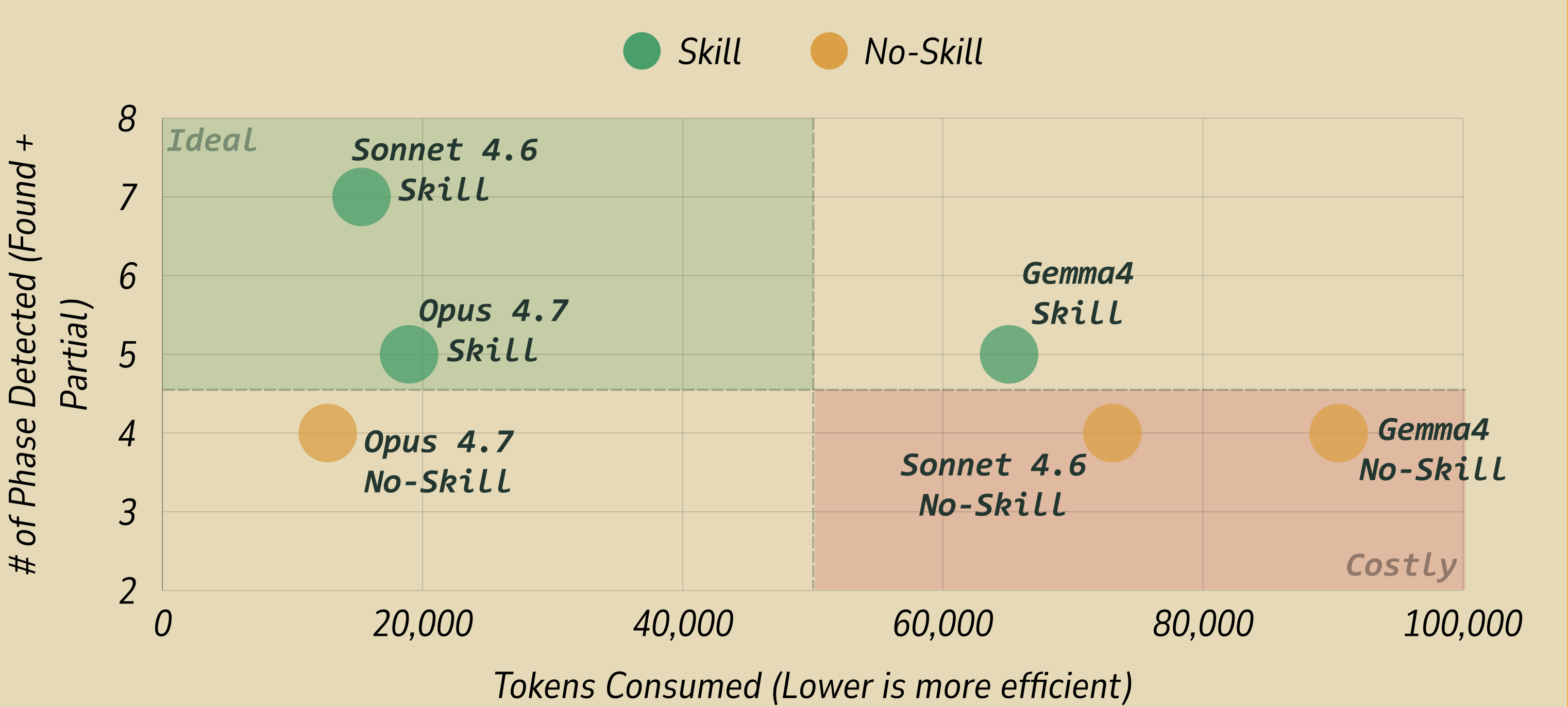
Does a skill-equipped smaller model outperform a more capable model operating without one – no retraining, no vector database?
- 3

Does skill integration transfer to open-source models?

Attack Timeline Benchmark



Performance Summary



Key Takeaways

- 1

Skill integration **consistently improves** detection across all model families.
- 2

Skill augmentation **compensates for model size**. Smaller models with skills rival larger ones with no skill.
- 3

Open-source models **benefit** from skill integration, but at higher token cost.

Future Works

- 1

Benchmark Skill Integration against RAG and PEFT.
- 2

Assess whether skill integration explores complete forensic evidence chains.
- 3

Investigate how LLMs internalize and weight skill documents at inference time

References

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